

1. Metode Persegi titik tengah

$$\text{Dik: } f(x) = 5 + 2x^2$$

$$n = 4$$

$$a = 0 ; b = 4 ;$$

$$\begin{aligned} x_1 &= a + (i - 1/2) \Delta x \\ &= 0 + (1 - 1/2) 1 \\ &= 0,5 \end{aligned}$$

$$\begin{aligned} x_3 &= a + (i - 1/2) \Delta x \\ &= 0 + (3 - 1/2) \Delta x \\ &= 2,5 \end{aligned}$$

$$\begin{aligned} f(x_1) &= 5 + 2x^2 \\ &= 5 + 2(0,5)^2 \\ &= 5 + 2(0,25) \\ &= 5 + 0,5 \\ &= 5,5 \end{aligned}$$

$$\begin{aligned} f(x_3) &= 5 + 2x^2 \\ &= 5 + 2(2,5)^2 \\ &= 5 + 2(6,25) \\ &= 5 + 12,5 \\ &= 17,5 \end{aligned}$$

$$\begin{aligned} W &= \Delta x [f(x_1) + f(x_2) + f(x_3) + f(x_4)] \\ &= 1 [5,5 + 9,5 + 17,5 + 29,5] \\ &= 1 \cdot 62 \\ &= 62 \text{ J } || \end{aligned}$$

$$\Delta x = \frac{4-0}{4} = 1 \rightarrow \Delta x = \frac{b-a}{n}$$

$$\begin{aligned} x_2 &= a + (i - 1/2) \Delta x \\ &= 0 + (2 - 1/2) 1 \\ &= 1,5 \end{aligned}$$

$$\begin{aligned} x_3 &= a + (i - 1/2) \Delta x \\ &= 0 + (4 - 1/2) 1 \\ &= 3,5 \end{aligned}$$

$$\begin{aligned} f(x_2) &= 5 + 2x^2 \\ &= 5 + 2(1,5)^2 \\ &= 5 + 2(2,25) \\ &= 5 + 4,5 \\ &= 9,5 \end{aligned}$$

$$\begin{aligned} f(x_4) &= 5 + 2x^2 \\ &= 5 + 2(3,5)^2 \\ &= 5 + 2(12,25) \\ &= 5 + 24,5 \\ &= 29,5 \end{aligned}$$

2. Metode Trapesium

$$\text{Dik: } C(T) = 20 + 0,05T^2$$

$$a = 0 ; b = 100 ; n = 4$$

$$\Delta T = \frac{100-0}{4} = 25$$

$$\begin{aligned} T_1 &= a + i \cdot \Delta T \\ &= 0 + 0 \cdot 25 \\ &= 0 \end{aligned}$$

$$\begin{aligned} T_3 &= a + i \cdot \Delta T \\ &= 0 + 2 \cdot 25 \\ &= 50 \end{aligned}$$

$$\begin{aligned} T_4 &= a + i \cdot \Delta T \\ &= 0 + 4 \cdot 25 \\ &= 100 \end{aligned}$$

$$\begin{aligned} T_2 &= a + i \cdot \Delta T \\ &= 0 + 1 \cdot 25 \\ &= 25 \end{aligned}$$

$$\begin{aligned} T_4 &= a + i \cdot \Delta T \\ &= 0 + 3 \cdot 25 \\ &= 75 \end{aligned}$$

$$\begin{aligned}
 C(T_1) &= 20 + 0,05 (0)^2 \\
 &= 20 + 0 \\
 &= 20
 \end{aligned}$$

$$\begin{aligned}
 C(T_2) &= 20 + 0,05 (25)^2 \\
 &= 20 + 0,05 \cdot 625 \\
 &= 20 + 31,25 \\
 &= 51,25
 \end{aligned}$$

$$\begin{aligned}
 C(T_3) &= 20 + 0,05 (50)^2 \\
 &= 20 + 0,05 \cdot 2500 \\
 &= 20 + 125 \\
 &= 145
 \end{aligned}$$

$$\begin{aligned}
 C(T_4) &= 20 + 0,05 (75)^2 \\
 &= 20 + 0,05 \cdot 5625 \\
 &= 20 + 281,25 \\
 &= 301,25
 \end{aligned}$$

$$\begin{aligned}
 C(T_5) &= 20 + 0,05 (100)^2 \\
 &= 20 + 0,05 \cdot 10000 \\
 &= 20 + 500 \\
 &= 520
 \end{aligned}$$

$$\begin{aligned}
 Q &= \frac{\Delta T}{2} [C(T_0) + 2C(T_1) + 2C(T_2) + 2C(T_3) + C(T_4)] \\
 &= \frac{25}{2} [20 + 2 \cdot 51,25 + 2 \cdot 145 + 2 \cdot 301,25 + 520] \\
 &= 12,5 [20 + 102,5 + 290 + 602,5 + 520] \\
 &= 12,5 \cdot 1535 \\
 &= 19.187,50 //
 \end{aligned}$$

3. Methode Simpson $\frac{1}{3}$

Dik: $I(t) = 2 + 3t + t^2$

$a = 0$; $b = 6$; $n = 6$; $\Delta t = 1$

$$\begin{aligned}
 t_0 &= a + i(\Delta t) \\
 &= 0 + 0 \cdot 1 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 t_1 &= a + i \Delta t \\
 &= 0 + 1 \cdot 1 \\
 &= 1
 \end{aligned}$$

$$\Delta t = \frac{6 - 0}{6} = 1$$

$$\begin{aligned}
 t_2 &= a + i \Delta t \\
 &= 0 + 2 \cdot 1 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 t_3 &= a + i \Delta t \\
 &= 0 + 3 \cdot 1 \\
 &= 3
 \end{aligned}$$

$$\begin{aligned}
 t_4 &= a + i \Delta t \\
 &= 0 + 4 \cdot 1 \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 t_5 &= a + i \Delta t \\
 &= 0 + 5 \cdot 1 \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 t_6 &= a + i \Delta t \\
 &= 0 + 6 \cdot 1 \\
 &= 6
 \end{aligned}$$

$$\begin{aligned}
 I(t_0) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 0 + 0^2 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 I(t_1) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 1 + 1^2 \\
 &= 6
 \end{aligned}$$

$$\begin{aligned}
 I(t_2) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 2 + 2^2 \\
 &= 12
 \end{aligned}$$

$$\begin{aligned}
 I(t_3) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 3 + 3^2 \\
 &= 20
 \end{aligned}$$

$$\begin{aligned}
 I(t_4) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 4 + 4^2 \\
 &= 30
 \end{aligned}$$

$$\begin{aligned}
 I(t_5) &= 2 + 3t + t^2 \\
 &= 2 + 3 \cdot 5 + 5^2 \\
 &= 42
 \end{aligned}$$

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$$\begin{aligned} I(t_6) &= 2 + 3t + t^2 \\ &= 2 + 3 \cdot 6 + 6^2 \\ &= 50 \end{aligned}$$

$$q_h = \frac{\Delta t}{3} [I(t_0) + 4I(t_1) + 2I(t_2) + 4I(t_3) + 2I(t_4) + 4I(t_5) + I(t_6)]$$

$$= \frac{1}{3} [2 + 4 \cdot 6 + 2 \cdot 12 + 4 \cdot 20 + 2 \cdot 30 + 4 \cdot 42 + 50]$$

$$= \frac{1}{3} [2 + 24 + 24 + 80 + 60 + 168 + 50]$$

$$= \frac{1}{3} \cdot 414$$

$$= 138 \text{ €}$$

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